Loneliness and nervousness mediate the association between alcohol use and cyberbullying victimization among school-aged adolescents: a two-year longitudinal study

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Resumo (Abstract)

Background Adolescent alcohol use and cyberbullying victimization represent significant global public health concerns. This two-year longitudinal study examined the prospective association between alcohol use and cyberbullying victimization among school-aged adolescents and explored the potential mediating roles of loneliness, sadness, and nervousness. Methods We utilized data from 1,902 adolescents (mean age 12.2 ± 0.47 years; 51.3% boys) in the Ningbo Youth Risk Behavior Survey (2016–2018). Self-reported baseline data included alcohol use, lifestyle factors (physical activity, screen time, smoking status, and diet), and demographics. Cyberbullying victimization was assessed at follow-up. Multivariable logistic regression, adjusted for key covariates, was used to evaluate the association between alcohol and cyberbullying. Sensitivity analyses were performed to test the consistency of this association across subgroups stratified by sex, age of drinking initiation, and alcohol consumption frequency. Subgroup and mediation analyses were performed to examine potential effect modifiers and the indirect effects of loneliness, sadness, and nervousness. Results At baseline, 40.6% (n = 773) of the adolescents reported alcohol use. The incidence of cyberbullying victimization at follow-up was 11.4% (n = 217). After adjusting for confounders, baseline alcohol use was significantly associated with increased odds of subsequent cyberbullying victimization (adjusted odds ratio [AOR] = 1.45, 95% CI: 1.06 to 2.00). Sensitivity analyses revealed that this association was stronger in boys (AOR = 1.63, 95% CI: 1.10 to 2.42), those who initiated drinking at age ≥ 11 years (AOR = 1.80, 95% CI: 1.23 to 2.61), and those with higher drinking frequency (1–2 days/month: AOR = 1.71, 95% CI: 1.09 to 2.63; 3–30 days/month: AOR = 2.41, 95% CI: 1.24 to 4.50). Mediation analyses indicated that loneliness and nervousness partially mediated the relationship between alcohol use and cyberbullying victimization, accounting for 11.59% and 7.25% of the total effect, respectively. Sadness did not have a significant mediating effect. Conclusions Alcohol use is an independent predictor of subsequent cyberbullying victimization among adolescents, with a more pronounced effect among boys. This association is partially mediated by increased feelings of loneliness and nervousness. These findings suggest that interventions targeting alcohol use, in addition to efforts to reduce loneliness and nervousness, may help mitigate the risk of cyberbullying victimization in this population.

Introduction

Adolescent alcohol use and cyberbullying victimization are pressing global public health challenges, each of which is associated with significant adverse outcomes for mental health, social development, and long-term well-being [1 , 2]. While a body of research has established a association between these two risk behaviors, a critical and unresolved question persists: what are the potential psychological processes that explain why adolescents who use alcohol are more likely to become victims of cyberbullying? Existing evidence is methodologically limited. Cross-sectional designs preclude causal inference, and many studies fail to adequately control for key confounders such as screen time, physical activity, and concurrent emotional states [3 , 4]. Furthermore, findings

regarding the role of sex and developmental stage have been inconsistent, with some studies reporting significant associations only in early adolescence or specific subgroups [5 – 7].

The investigation of mediating psychological candidate mechanisms offers a promising avenue to unravel this complexity. Theoretical frameworks, such as the general aggression model, posit that internalizing symptoms can serve as critical potential pathways linking adverse experiences to subsequent maladaptive behaviors [8 , 9]. In the context of alcohol use and victimization, specific emotional states—loneliness, sadness, and nervousness—are compelling candidate mediators. Loneliness, sadness, and nervousness were selected as distinct candidate mediators within the broader domain of internalizing symptoms, given their established links to both alcohol use and peer victimization in prior literature [8 – 12]. Loneliness, or perceived social isolation, is a well-documented consequence of both problematic alcohol use and peer victimization, and it can exacerbate social anxiety and impair conflict-resolution skills, potentially increasing vulnerability to online aggression. Nervousness, reflecting acute situational worry and anxiety, is closely tied to alcohol use as both a potential motivator for drinking (self-medication) and an outcome of stressful victimization experiences [10 , 11]. Sadness, or a dysphoric mood, has been linked to cyberbullying outcomes, although its distinct role is often conflated with broader emotional distress [8 , 12]. Examining specific mediators is theoretically important because internalizing symptoms may serve as key processes linking alcohol use to increased vulnerability to online victimization, as posited by frameworks such as the general aggression model. This approach can also inform targeted interventions by identifying modifiable psychological factors.

However, significant gaps remain in the literature. First, most studies have examined these psychological factors as outcomes or correlates rather than formally testing their mediating role in a longitudinal, alcohol-to-cyberbullying framework. Second, research often aggregates these distinct emotional constructs under broad labels such as “psychological distress” or “internalizing problems,” obscuring their unique contributions [13 , 14]. Third, there is a paucity of longitudinal studies that simultaneously adjust for a comprehensive set of lifestyle and demographic confounders, which is essential to isolate the specific effect of alcohol use and the proposed mediators [13]. Consequently, the dose■response relationship between alcohol use patterns (e.g., initiation age and frequency) and cyberbullying risk and the extent to which loneliness, sadness, and nervousness independently explain this link remain quantitatively underexplored.

To address these critical gaps, the present study employs a two-year longitudinal design with data from a large school-based cohort of Chinese adolescents. Our research is strategically designed to achieve three sequential objectives: (1) to establish a prospective association between baseline alcohol use and subsequent cyberbullying victimization, adjusting for a wide array of potential confounders; (2) to conduct sensitivity and subgroup analyses to examine the consistency of this association across sex, age of drinking initiation, and drinking frequency; and (3) to formally test the independent and specific mediating roles of loneliness, sadness, and nervousness in this association using established mediation analytical techniques. Our results provide a scientific basis for the development of effective preventive and intervention measures to protect the physical and mental health of adolescents and promote their overall development through an in-depth analysis of how these factors are interrelated with adolescents’ alcohol use and cyberbullying victimization.

Methods

Study design and participants

This secondary analysis utilized data from the Ningbo Youth Risk Behavior Survey (NYRBS), a school-based prospective study conducted from 2016 to 2018 in Ningbo, Zhejiang Province, China. The study employed a multi-stage cluster sampling design. At baseline (September-November 2016), 13 middle schools were purposively selected from across 10 administrative districts. All seventh-grade students from the selected schools were invited to participate. Follow-up assessments were conducted in October 2017 and October 2018. The cohort comprised generally healthy adolescents; individuals with major pre-existing illnesses or those providing incomplete data were excluded from the analytical sample.

Trained personnel administered a structured, self-report questionnaire during regular class hours, adapted from the U.S. Youth Risk Behavior Survey (YRBS) [15 – 17]. The protocol was approved by the Institutional Review Board of the Ningbo Center for Disease Control and Prevention (Approval No. 201703). Prior to participation, written informed consent was obtained from a parent or legal guardian for each student, with student assent also secured. This study adheres to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines for cohort studies. A participant flow diagram is provided in Fig. 1 .

Fig. 1 Flow chart of this study

Measures

Demographic and lifestyle covariates

Demographic information, including age, sex, ethnicity, parental marital status, and parental education level, was collected at baseline [18]. Lifestyle behaviors assessed at baseline included physical activity, screen time, cigarette use, and dietary habits [19 , 20]. Physical activity was categorized as engaging in moderate-to-vigorous or muscle-strengthening activities at least once per week. Excessive screen time was defined as > 2 h per day of television viewing or recreational internet use [16 , 17]. Cigarette use was coded positively if a student reported smoking at least once in the preceding month. Dietary intake was assessed via frequency questions; for analysis, regular fruit and meat consumption were dichotomized (≥ 1 time/day vs. less) [20].

Primary exposure: alcohol use

Alcohol use was the primary exposure variable, measured at baseline (2016). Participants were asked about their drinking behavior in the past month. Alcohol use behavior was assessed through the following question: “Have you ever drunk alcohol?” (1 = Yes, 0 = No). To further characterise alcohol consum.ption, we asked participants their age at first drink (“How old were you when you first drank alcohol?”) and the number of days they drank in the past 30 days (“On how many days did you consume alcohol in the past 30 days?”). Primary analyses treated ever-drinking as a binary variable, while sensitivity analyses used categories based on age at initiation and monthly drinking days.

Outcome: cyberbullying victimization

The outcome, cyberbullying victimization, was assessed during the two follow-up waves (2017, 2018) using a single item: “During the past 12 months, have you been electronically bullied (e.g., via email, chat rooms, instant messaging, websites, or texting)?” (Response: Yes/No) [21 , 22]. Cyberbullying victimization was defined as a cumulative (Yes/No) outcome (victimization reported in 2017 or 2018 or both). The internal consistency (Cronbach’s α) for bullying related items (including related three items-cyberbullying victimization, school bullying victimization and school bullying perpetrator, the

late two item was not reported in this manuscript) in this sample was 0.738.

Candidate mediators: psychological factors

Potential mediating psychological factors—loneliness, sadness, and nervousness—were assessed at the first follow-up (Year 1, 2017) [23]. Loneliness was evaluated on a 5-point frequency scale (0 = never to 4 = always). Sadness and nervousness were dichotomous (Yes/No) variables, capturing sustained feelings of sadness/hopelessness (≥ 2 weeks) or nervousness/worry (≥ 1 week), respectively, that disrupted daily activities [24].

Statistical analysis

Descriptive statistics were computed for all the variables. Continuous variables are presented as mean ± standard deviation (SD) or median (interquartile range, IQR) based on their distribution, assessed via the Kolmogorov-Smirnov test. Categorical variables are summarized as frequencies (percentages).

Between-group differences for continuous variables were evaluated using independent t tests or one-way analysis of variance (ANOVA), as appropriate. Categorical variables were compared using chi-square tests.

Multivariate logistic regression models were applied to examine the association between alcohol use and cyberbullying victimization, with the results reported as adjusted odds ratios (ORs) and 95% confidence intervals (CIs). The models were adjusted for the following covariates: age, sex, parental marital status, parental education at the college level, moderate physical activity, outdoor activity, television viewing, computer use, cigarette use, fruit intake, vegetable intake, meat consumption, sadness, and nervousness.

To further investigate the dose■response relationship between alcohol use and cyberbullying victimization, we conducted three sensitivity analyses. First, participants were stratified by sex (boys and girls), and separate univariate and multivariable models were fitted within each subgroup. Second, on the basis of age at drinking initiation, participants were categorized into three groups: never drinkers, those who began drinking before age 11, and those who started drinking at age 11 or older. Univariate and multivariable models were constructed for each subgroup. Third, according to the number of drinking days per month, participants were classified into nondrinkers, those who drank 1–2 days per month, and those who drank 3–30 days per month, with separate univariate and multivariable models applied.

Subgroup analyses were performed to assess potential effect modification by sex, cigarette use, moderate physical activity, outdoor activity, television viewing, computer use, meat intake, sadness, and nervousness. Additionally, a mediation analysis was conducted to examine whether loneliness, sadness, or nervousness indirectly influenced the association between alcohol use and cyberbullying victimization.

All analyses were performed using R software version 5.3.0 (R Foundation for Statistical Computing, Vienna, Austria). Statistical tests were two-sided, and a P value < 0.05 was considered to indicate statistical significance.

Results

Descriptive analysis

A total of 1902 adolescents were included in our study. Information on the characteristics of the study sample divided according to alcohol use is shown in Table 1 . A total of 773 (40.6%) adolescents reported engaging in alcoholic behavior. The average age was 12.2 ± 0.47 years. Most participants were boys (51.3%) or Han Chinese (98.1%). Most participants' parents were married (92.6%), and the participants' parents did not have a college degree (77.2%). Compared with the nonalcohol-use group, the alcohol-use group was more significantly associated with less moderate physical activity, more time watching television and using the computer, more frequent fast food and meat intake, less fish intake, a greater likelihood of smoking and fighting, and a greater likelihood of feeling sad and nervous (all P values < 0.05).

Table 1 Characteristics of the study subjects according to alcohol use (n = 1902) Characteristics Total (n = 1902) Nonalcohol use(n = 1129) Alcohol use(n = 773) P value Age, years, mean ± SD 12.2 ± 0.47 12.2 ± 0.46 12.2 ± 0.49 0.481 Gender, n (boy, %) 975 (51.3) 547 (48.4) 428 (55.4) 0.004 Nationality (Han, %) 1865 (98.1) 1103 (97.7) 762 (98.6) 0.232 Parental marriage status, n (married, %) 1739 (92.6) 1038 (93.1) 701 (91.9) 0.367 Parental education at the college level, n (%) 0.193 Both had a college degree 254 (14.2) 161 (15.0) 93 (12.9) Only one of them had a college degree 244 (13.6) 154 (14.3) 90 (12.5) None of them had a college degree 1296 (72.2) 759 (70.7) 537 (74.6) Physical activity, n (%) Moderate physical activity (≥ 1 day/week) 1758 (92.5) 1061 (94.1) 697 (90.2) 0.002 Muscle strengthening activity (≥ 1 day/week) 1165 (61.3) 702 (62.2) 463 (59.9) 0.327 Outdoor activity (≥ 1 day/week) 1686 (88.7) 1006 (89.2) 680 (88.1) 0.502 Sedentary activity, n (%) Watch television (≥ 2 h/day) 448 (23.6) 215 (19.0) 233 (30.1) < 0.001 Use computer (≥ 2 h/day) 379 (19.9) 186 (16.5) 193 (25.0) < 0.001 Dietary behaviors, n (%) Fruits (≥ 1 time/day) 832 (43.7) 494 (43.8) 338 (43.7) 1.000 Vegetables (≥ 2 times/day) 1684 (88.6) 1009 (89.4) 675 (87.4) 0.219 Fast food (≥ 1 time/week) 909 (47.8) 515 (45.6) 394 (51.0) 0.024 Water (≥ 4 cups/day) 1156 (60.8) 696 (61.6) 460 (59.6) 0.392 Meat (≥ 1 time/day) 707 (37.3) 395 (35.1) 312 (40.6) 0.018 Fish (≥ 1 time/week) 1729 (90.9) 1039 (92.0) 690 (89.3) 0.048 Milk (≥ 1 time/day) 858 (45.1) 506 (44.8) 352 (45.5) 0.793 Other behaviors, n (%) Cigarette use (yes) 127 (6.68) 29 (2.57) 98 (12.7) < 0.001 Sadness (yes) 274 (14.4) 124 (11.0) 150 (19.4) < 0.001 Nervousness (yes) 599 (31.5) 306 (27.1) 293 (37.9) < 0.001

Incidence of cyberbullying victimization during follow-up

Information on the characteristics of the study sample, divided according to cyberbullying victimization, is shown in Table 2 . A total of 217 (11.4%) of the adolescents had engaged in cyberbullying victimization. Adolescents with electronic bullying, whose parents were more likely to be married, have more time using the computer, consume more fast food and meat, have less fish intake, be more prone to smoke and alcohol abuse, have more fights, and be more likely to perceive sadness and nervousness (all P values < 0.05).

Table 2 Characteristics of the study subjects according to cyberbullying victimization status (n = 1902) Characteristics Total (n = 1902) Noncyberbulling Victimization (n = 1685) Cyberbulling Victimization (n = 217) P value Age, years, mean ± SD 12.2 ± 0.47 12.2 ± 0.46 12.2 ± 0.54 0.836 Gender, n (boy, %) 975 (51.3) 824 (48.9) 151 (69.6) < 0.001 Nationality (han, %) 1865 (98.1) 1654 (98.2) 211 (97.2) 0.304 Parental marriage status, n (married, %) 1739 (92.6) 1552 (93.3) 187 (87.0) 0.001 Parental education at the college level, n (%) 0.378 Both had a college degree 254 (14.2) 228 (14.3) 26 (12.7) Only one of them had a college degree 244 (13.6) 210 (13.2) 34 (16.6) None of them had a college degree 1296 (72.2) 1151 (72.4) 145 (70.7) Physical activity, n (%) Moderate physical activity (≥ 1 day/week) 1758 (92.5) 1562 (92.8) 196 (90.3) 0.253 Muscle strengthening activity (≥ 1 day/week) 1165 (61.3) 1044 (62.0) 121 (55.8) 0.089 Outdoor activity (≥ 1 day/week) 1686 (88.7) 1497 (88.9) 189 (87.1) 0.485 Sedentary activity, n (%) Watch television (≥ 2 h/day) 448 (23.6) 385

(22.8) 63 (29.0) 0.053 Use computer (≥ 2 h/day) 379 (19.9) 322 (19.1) 57 (26.3) 0.017 Dietary behaviors, n (%) Fruits (≥ 1 time/day) 832 (43.7) 751 (44.6) 81 (37.3) 0.051 Vegetables (≥ 2 times/day) 1684 (88.6) 1495 (88.7) 189 (87.5) 0.675 Fast food (≥ 1 time/week) 909 (47.8) 788 (46.8) 121 (55.8) 0.015 Water (≥ 4 cups/day) 1156 (60.8) 1024 (60.8) 132 (60.8) 1.000 Meat (≥ 1 time/day) 707 (37.3) 609 (36.3) 98 (45.6) 0.001 Fish (≥ 1 time/week) 1729 (90.9) 1541 (91.5) 188 (86.6) 0.028 Milk (≥ 1 time/day) 858 (45.1) 762 (45.2) 96 (44.2) 0.84 Other behaviors, n (%) Cigarette use (yes) 127 (6.68) 102 (6.05) 25 (11.5) 0.004 Alcohol use (yes) 773 (40.6) 653 (38.8) 120 (55.3) 0.001 Sadness (yes) 274 (14.4) 213 (12.6) 61 (28.1) 0.001 Nervousness (yes) 599 (31.5) 505 (30.0) 94 (43.3) 0.001

Associations between alcohol use and cyberbullying victimization

The relationships between sex, alcohol age, frequency of alcohol use, and cyberbullying victimization are shown in Table 3 . Overall, the effects of alcohol use, sex, alcohol age, and frequency of alcohol use on cyberbullying victimization were statistically significant ($P < 0.05$). Adjustments were made for age; sex; parental marital status; parental education at the college level; moderate physical activity; outdoor activity; watching television; and the use of computers, fruits, vegetables, meat, sadness, nervousness, and cyberbullying victimization, which were associated with an increased likelihood of alcohol use (AOR = 1.45, 95% CI = 1.06 to 2.00; $P < 0.05$). In addition, a multiadjusted model revealed that cyberbullying victimization was associated with an increased risk for boys (AOR = 1.63; 95% CI = 1.10 to 2.42; $P < 0.05$), those with an alcohol age ≥ 11 years (AOR = 1.80; 95% CI = 1.23 to 2.61; $P < 0.05$), and those who drank more than 1 to 2 days and 3 to 30 days in a month (AOR = 1.71; 95% CI = 1.09 to 2.63; $P < 0.05$; and AOR = 2.41; 95% CI = 1.24 to 4.50; $P < 0.05$, respectively).

Table 3 Association between alcohol use and cyberbullying victimization Alcohol Use Cyberbulling Victimization Crude Multi-adjusted # No Yes OR (95%CI) P value OR (95%CI) P value Overall(n = 1902) No 1032 97 1.0 (ref) 1.0 (ref) Yes 653 120 1.96(1.47–2.60) < 0.001 1.45(1.06-2.00) 0.021 Boys(n = 975) No 483 64 1.0 (ref) 1.0 (ref) Yes 341 87 1.93(1.36–2.74) < 0.001 1.63(1.10–2.42) 0.014 Girls(n = 927) No 549 33 1.0 (ref) 1.0 (ref) Yes 312 33 1.76(1.06–2.91) 0.0274 1.15(0.64–2.02) 0.628 Alcohol age(n = 1902) never alcohol 1015 91 1.0 (ref) 1.0 (ref) < 11 years 337 65 2.15(1.52–3.02) < 0.001 1.39(0.94–2.04) 0.098 ≥ 11 years 333 61 2.04(1.44–2.88) < 0.001 1.80(1.23–2.61) 0.002 Frequency of alcohol use(n = 1901) 0d 1481 159 1.0 (ref) 1.0 (ref) 1–2 d 154 39 2.36(1.58–3.45) < 0.001 1.71(1.09–2.63) 0.016 3–30 d 49 19 3.61(2.03–6.19) < 0.001 2.41(1.24–4.50) 0.007 # Adjustments for age, sex, parental marital status, parental education at the college level, moderate physical activity, outdoor activity, watching television, use of a computer, cigarette use, fruits, vegetables, meat, sadness, and nervousness

Subgroup analyses confirmed the above findings (Table 4). After adjustments were made for age, sex, parental marital status, parental education at the college level, moderate physical activity, outdoor activity, watching television, using a computer, cigarette use, fruits, vegetables, meat, sadness, nervousness, and alcohol use, these variables were determined to be related to sex, smoking status, meat, sadness, and nervousness (P for trend < 0.05). Stratified analyses according to potential risk factors for alcoholism, including sex, cigarette use, moderate physical activity, outdoor activity, watching television, using computers, meat, sadness, and nervousness, were performed. We found a positive correlation between alcohol use and cyberbullying victimization across the subgroups. However, there was no interaction effect between alcohol use and any of these factors except sex (P for interaction < 0.001).

Table 4 Subgroup analyses of the association between alcohol use and cyberbullying victimization Subgroups Noncyberbulling Victimization Cyberbulling Victimization # P P trend P interaction Gender < 0.001 < 0.001 Boys 1(ref) 1.63(1.10–2.42) 0.014 Girls 1(ref) 1.15(0.65–2.02) 0.628 Cigarette use 0.048 0.886 No 1(ref) 1.46(1.05–2.03) 0.025 Yes 1(ref) 1.20(0.03–5.80) 0.804 Moderate physical activity 0.358 0.127 <1 day/week 1(ref) 5.67(1.52–27.1) 0.015 ≥ 1 day/week 1(ref) 1.36(0.97–1.89) 0.074 Outdoor activity 0.464 0.684 <1 day/week 1(ref) 1.25(0.46–3.38) 0.663 ≥ 1 day/week 1(ref) 1.47(1.04–2.07) 0.027 Watch television 0.158 0.085 <2 h/day 1(ref) 1.75(1.21–2.55) 0.003 ≥ 2 h/day 1(ref) 0.93(0.50–1.72) 0.819 Use computer 0.050 0.634 <2 h/day 1(ref) 1.38(0.96–1.99) 0.083 ≥ 2 h/day 1(ref) 1.53(0.77–3.09) 0.228 Meat intake 0.017 0.995 <1 time/day 1(ref) 1.51(0.99–2.29) 0.055 ≥ 1 time/day 1(ref) 1.38(0.84–2.28) 0.201 Sadness < 0.001 0.176 No 1(ref) 1.30(0.90–1.88) 0.164 Yes 1(ref) 1.90(0.96–3.91) 0.072 Nervousness < 0.001 0.553 No 1(ref) 1.57(1.04–2.36) 0.032 Yes 1(ref) 1.30(0.78–2.19) 0.314 # The analyses were adjusted for age; sex; parental marital status; parental education at the college level; moderate physical activity; outdoor activity; watching television; and the use of computers, cigarettes, fruits, vegetables, meat, sadness, and nervousness when they were not the strata variables

Mediation analysis

Mediation analysis was conducted to explore the roles of loneliness, sadness, and nervousness in the relationship between alcohol use and cyberbullying victimization (Table 5 ; Fig. 2). Loneliness partially mediated the relationship between alcohol use and cyberbullying victimization, accounting for 11.59% of the total effect (indirect effect β = 0.008, P = 0.004; Table 5). Similarly, nervousness exhibited a partially mediated effect, contributing to 7.25% of the association (indirect effect β = 0.005, P = 0.022; Table 5). In contrast, sadness had no significant mediating effect (P = 0.317), suggesting that the link between alcohol use and cyberbullying victimization is partly explained by increased loneliness and nervousness.

Table 5 Mediation analysis of psychological factors (loneliness and nervousness) Type of Effect Mediation Pathway Effect Size SE Bootstrap 95% CI P Value Ratio of Effect Direct effect Alcohol use→cyberbullying victimization 0.056 0.015 0.027, 0.088 < 0.001 81.16% Mediation Effect Alcohol use→loneliness→cyberbullying victimization 0.008 0.003 0.003, 0.015 0.004 11.59% Alcohol use→nervousness→cyberbullying victimization 0.005 0.002 0.002, 0.010 0.022 7.25% Total Mediation Effect 0.013 0.004 0.007, 0.021 < 0.001 18.84% Total effect 0.069 0.015 0.040, 0.102 < 0.001

Fig. 2 Multiple parallel longitudinal associations between alcohol use and cyberbullying victimization are mediated by loneliness and nervousness

Discussion

Principal findings

This two-year longitudinal study of 1,902 Chinese school-aged adolescents provides robust evidence that baseline alcohol use is prospectively associated with an increased risk of cyberbullying victimization at follow-up, independent of a comprehensive set of demographic, lifestyle, and behavioral confounders. After full adjustment, adolescents reporting alcohol use at baseline had 45% higher odds of subsequent cyberbullying victimization (AOR = 1.45, 95% CI: 1.06–2.00). Sensitivity analyses further demonstrated a dose–response pattern, with stronger associations among boys (AOR = 1.63), those who reported initiating drinking at age ≥ 11 years (AOR = 1.80), and those with

higher drinking frequency (up to AOR = 2.41 for 3–30 days/month). Mediation analyses revealed that loneliness (11.59% of the total effect) and nervousness (7.25% of the total effect) were significant partial mediators, whereas sadness had no mediating role. These findings directly address the gaps highlighted in the Introduction by establishing temporal precedence, rigorously controlling for confounders, and disaggregating the specific contributions of distinct emotional states—loneliness and nervousness—in a longitudinal framework.

Interpretation of findings and plausible mechanisms

The prospective link between adolescent alcohol use and later cyberbullying victimization may extends prior cross-sectional and short-term reciprocal evidence, most of which has emphasized victimization as a predictor of substance use rather than the reverse direction examined here [1 , 2 , 13]. Alcohol consumption during early adolescence often occurs in social or online contexts that heighten exposure to unstructured peer interactions and reduce inhibitory control, thereby increasing vulnerability to online aggression [25 , 26]. Possible explanations consistent with prior literature include impaired judgment and impulsivity associated with alcohol use, which may increase exposure to riskier online behaviors—such as sharing personal information or frequenting less moderated digital spaces—making them more visible targets for cyberbullying [27 , 28]. Furthermore, alcohol-related emotional dysregulation may further perpetuate a cycle of social withdrawal or conflict that manifests in both offline and online environments [2 , 29 , 30].

The observed dose–response relationship (stronger effects with earlier initiation and greater frequency) may reinforces alcohol use as a behavioral marker of underlying vulnerability rather than a mere correlate. Early and frequent drinking may reflect preexisting traits such as impulsivity or affiliation with deviant peers, which in turn amplify online risk exposure and social marginalization [31]. This graded pattern may strengthens the case for a potential causal contribution, which is consistent with theoretical models such as the general aggression model, which posits that internalizing states and behavioral disinhibition link adverse experiences to subsequent victimization.

Specific mediating roles of loneliness and nervousness

A key innovation of this study is the formal mediation analysis demonstrating that loneliness and nervousness—assessed at the intermediate time point—partially explain the alcohol–cyberbullying pathway. Loneliness accounted for 11.59% of the total effect, suggesting that alcohol use may exacerbate perceived social isolation (or share common antecedents such as poor social skills), prompting adolescents to seek connections in online spaces where they are less equipped to detect or deflect aggression. This mechanism aligns with evidence that loneliness heightens vulnerability to cyberbullying by impairing conflict-resolution skills and increasing reliance on digital platforms for social compensation [32].

Similarly, nervousness (accounting for 7.25% of the effect) represents a pathway of heightened emotional reactivity. Alcohol may be used as a maladaptive coping mechanism for preexisting anxiety, or its use may induce or worsen nervousness. This heightened state of arousal and worry can impair social processing, potentially making adolescents appear more vulnerable or reactive, thereby attracting victimization [26 , 33].

The nonsignificant mediating role of sadness is noteworthy. Unlike loneliness (social pain) or nervousness (hypervigilance), sadness is typically characterized by anhedonia and withdrawal. It is plausible that sadness does not increase online visibility or perceived vulnerability in the same manner as the other two constructs, explaining its lack of mediation. Future studies should examine

specific subtypes of internalizing symptoms to confirm this distinction [32 , 33].

Subgroup differences and moderation by demographic and behavioral factors

Subgroup analyses revealed that the alcohol–cyberbullying association was consistently positive across most strata but significantly stronger among boys, which is consistent with prior reports of sex-specific patterns in cyberbullying risk and alcohol-related disinhibition [34]. Boys' greater involvement in risk-taking behaviors and visibility in certain online peer networks may amplify the pathway. In our study, boys, adolescents who started drinking earlier, and adolescents who drank more frequently were more likely to engage in cyberbullying [35]. No significant interactions emerged for most other lifestyle factors (e.g., smoking, physical activity, diet), suggesting that the association between alcohol use and cyberbullying is a robust phenomenon across diverse behavioral strata. This consistency implies that the underlying psychological processes (e.g., impaired judgment) may operate similarly regardless of physical activity or diet, highlighting the central role of emotional states identified in our mediation analysis [36].

Cyberbullying victimization has been associated with several adverse outcomes in adolescents, including insomnia, social anxiety, alcohol use, poor academic performance, and suicidal ideation [37]. Although there was a positive correlation between alcohol consumption and cyberbullying victimization in the majority of the subgroups, no significant interactions were detected between alcohol consumption and any other potential risk factor except sex.

Strengths and limitations

The strengths of this study include its longitudinal design, which supports temporal precedence, the adjustment for a comprehensive set of relevant confounders, and the formal testing of specific mediators. However, several limitations must be acknowledged. First, the sample was drawn from one region in China using convenience sampling, which may limit the generalizability of the findings to other cultural or national contexts. Future multicenter studies are needed. Second, all the measures were self-reported, which may introduce recall and social desirability biases. The incorporation of objective measures (e.g., biomarkers for alcohol and digital trace data) in future research would be valuable. Third, while we controlled for key covariates, residual confounding from unmeasured variables (e.g., parenting style, peer delinquency, genetic factors) is possible. Fourth, mediators were assessed at the first follow-up wave (2017), which overlaps with the first assessment of the outcome. Although this design supports temporal ordering for the second-wave outcome, we cannot fully rule out reverse causation or bidirectionality between mediators and cyberbullying victimization. Future studies with more frequent assessments are needed to clarify these dynamics. Fifth, several key variables were measured with limited precision. Alcohol use was assessed as a binary ever-use variable in the past month, without detailed quantity or problematic drinking measures. Cyberbullying victimization relied on a single self-reported dichotomous item, which does not capture frequency, severity, or duration and may underestimate variability. Mediators were partially dichotomized, potentially reducing nuance. Finally, the effect sizes for the mediated associations, while statistically significant, were modest, indicating that other important candidate mechanisms (e.g., impaired executive function and deviant peer affiliation) remain to be explored.

Implications and future directions

These findings have clear implications for prevention and intervention. They suggest that programs aimed at reducing cyberbullying victimization should consider screening for alcohol use as a risk indicator, particularly among boys. Interventions could be enhanced by simultaneously targeting the reduction in alcohol use and the development of socioemotional skills to counteract loneliness and manage nervousness. For instance, school-based programs that foster healthy peer connections and teach adaptive coping and anxiety-management strategies may help disrupt the identified pathway.

Future research should aim to replicate these findings in diverse populations, utilize more frequent assessment waves to better capture dynamic processes, and investigate the potential moderating role of factors such as peer group norms and parental monitoring. Examining the interplay between these affective mediators and other cognitive or social mechanisms will provide a more comprehensive understanding of the risk landscape for adolescent cyberbullying victimization.

Conclusion

In conclusion, this longitudinal study demonstrated that alcohol use in early adolescence is associated with a greater likelihood of subsequent cyberbullying victimization, with the relationship being stronger among boys and those with greater drinking frequency and earlier initiation. Importantly, the association is partially explained by increases in feelings of loneliness and nervousness. These findings underscore that the link between substance use and online victimization is not merely behavioral but is importantly may rooted through affective pathways. Moving forward, integrated prevention strategies that address underage drinking, promote healthy social integration, and develop emotional regulation skills may be effective at reducing the burden of cyberbullying among adolescents.